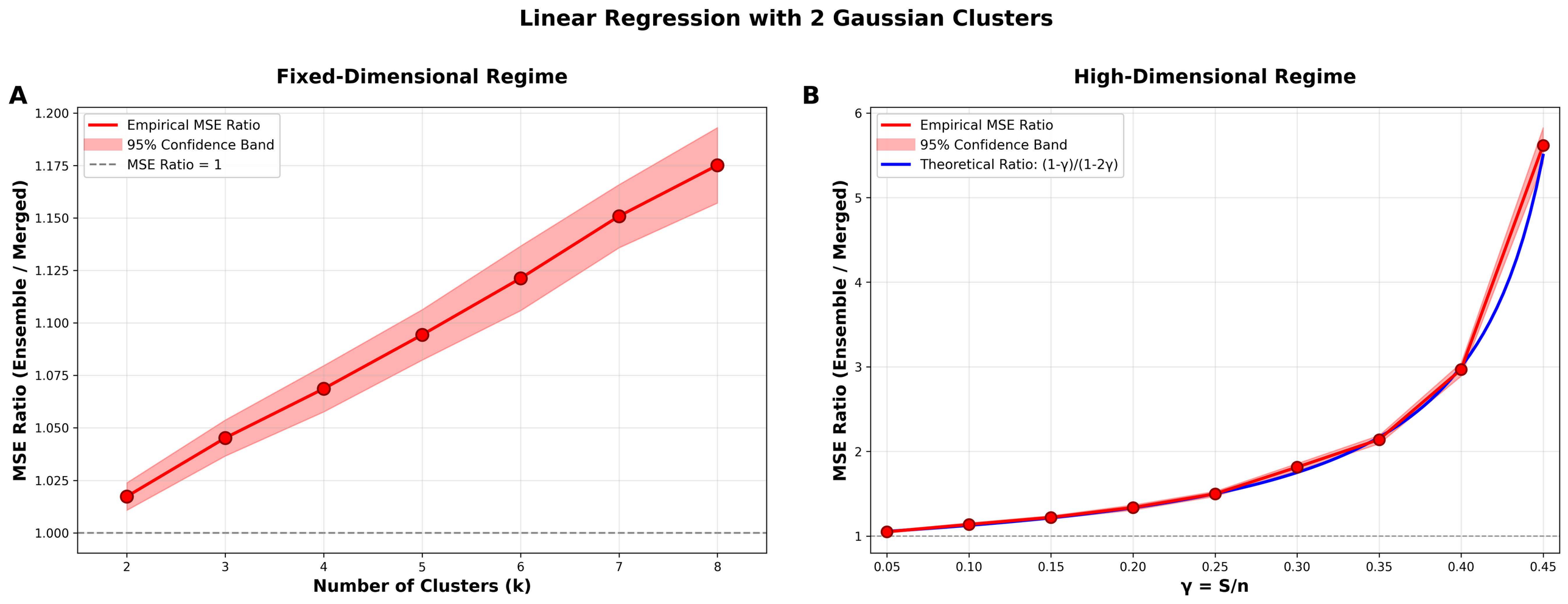

Linear Regression with 2 Gaussian Clusters
A
Fixed-Dimensional Regime
Empirical MSE Ratio
95% Confidence Band
MSE Ratio = 1
MSE Ratio (Ensemble / Merged)
Number of Clusters (k)
B
High-Dimensional Regime
Empirical MSE Ratio
95% Confidence Band
Theoretical Ratio: (1-γ)/(1-2γ)
MSE Ratio (Ensemble / Merged)
γ = S/n